

Wind Turbine Gearbox Predictive Maintenance: A Comprehensive Machine Learning and Deep Learning Approach for Anomaly Detection and Remaining Useful Life Prediction

Rüzgar Türbini Dişli Kutusu Kestirimci Bakım:
Anomali Tespiti ve Kalan Kullanım Ömrü Tahmini için
Kapsamlı Makine Öğrenmesi ve Derin Öğrenme Yaklaşımı

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Abstract

This study presents a comprehensive predictive maintenance framework for wind turbine gearboxes utilizing five years of SCADA (Supervisory Control and Data Acquisition) sensor data comprising 262,800 hourly records. The proposed approach integrates multiple machine learning paradigms including classical supervised learning (Random Forest, XGBoost, LightGBM, Logistic Regression), unsupervised anomaly detection (Isolation Forest, One-Class SVM, Local Outlier Factor, Autoencoder), deep learning architectures for time series analysis (LSTM, TCN, Transformer Encoder), and hybrid ensemble methods (Stacking, Weighted Voting). A total of 95 features were engineered from 7 original sensor measurements through rolling statistics, lag features, and Fourier transforms. The unsupervised methods achieved exceptional performance with One-Class SVM reaching 0.9999 ROC-AUC and 0.9965 PR-AUC. Among deep learning models, Temporal Convolutional Network (TCN) demonstrated the best balanced performance with F1-score of 0.445 and PR-AUC of 0.601. The study further extends beyond binary anomaly detection to Remaining Useful Life (RUL) prediction, where the LSTM regression model achieved a Mean Absolute Error of 3.96 hours with 100% early warning accuracy at 24, 48, and 72-hour horizons. Explainability analysis through SHAP and LIME provides interpretable insights into model decisions. The proposed framework demonstrates that combining supervised, unsupervised, and deep learning methods with temporal feature engineering enables highly effective predictive maintenance for wind turbine gearboxes, bridging the gap from anomaly detection to actionable maintenance scheduling.

Keywords: Predictive maintenance, wind turbine, gearbox, anomaly detection, remaining useful life, SCADA, deep learning, LSTM, TCN, Transformer, ensemble learning, SHAP, XGBoost, Random Forest

Özet

Bu çalışma, rüzgar türbini dişli kutuları için beş yıllık SCADA (Merkezi Denetim Kontrol ve Veri Toplama) sensör verisi kullanarak kapsamlı bir kestirimci bakım çerçevesi sunmaktadır. Toplam 262.800 saatlik kayıt içeren veri seti üzerinde, klasik denetimli öğrenme (Random Forest, XGBoost, LightGBM, Lojistik Regresyon), denetimsiz anomali tespiti (Isolation Forest, One-Class SVM, Local Outlier Factor, Autoencoder), zaman serisi derin öğrenme mimarileri (LSTM, TCN, Transformer Encoder) ve hibrit topluluk yöntemleri (Stacking, Ağırlıklı Oylama) olmak üzere çoklu makine öğrenmesi paradigmaları entegre edilmiştir. Orijinal 7 sensör ölçümünden yuvarlanan istatistikler, gecikme özellikleri ve Fourier dönüşümleri aracılığıyla toplam 95 özellik mühendisliği gerçekleştirilmiştir. Denetimsiz yöntemler arasında One-Class SVM, 0,9999 ROC-AUC ve 0,9965 PR-AUC değerleri ile olağanüstü performans göstermiştir. Derin öğrenme modelleri arasında Temporal Konvolüsyonel Ağ (TCN), 0,445 F1-skoru ve 0,601 PR-AUC ile en dengeli performansı sergilemiştir. Çalışma, ikili anomali tespitinin ötesine geçerek Kalan Kullanım Ömrü (RUL) tahminine genişletilmiş olup, LSTM regresyon modeli 3,96 saatlik Ortalama Mutlak Hata ve 24, 48, 72 saatlik ufuklarda %100 erken uyarı doğruluğu elde etmiştir. SHAP ve LIME yoluyla yapılan açıklanabilirlik analizi, model kararlarına ilişkin yorumlanabilir içgörüler sağlamaktadır. Önerilen çerçeve, denetimli, denetimsiz ve derin öğrenme yöntemlerinin zamansal özellik mühendisliği ile birleştirilmesinin, rüzgar türbini dişli kutuları için yüksek düzeyde etkili kestirimci bakımı mümkün kıldığını ve anomali tespitinden uygulanabilir bakım planlamasına köprü kurduğunu göstermektedir.

Anahtar Kelimeler: Kestirimci bakım, rüzgar türbini, dişli kutusu, anomali tespiti, kalan kullanım ömrü, SCADA, derin öğrenme, LSTM, TCN, Transformer, topluluk öğrenmesi, SHAP, XGBoost, Random Forest

I. INTRODUCTION

Wind energy has emerged as one of the most promising renewable energy sources worldwide, with installed capacity growing exponentially over the past two decades. Wind turbines, as the primary conversion devices, operate under harsh environmental conditions including variable wind speeds, temperature fluctuations, and mechanical stress. Among the critical components, the gearbox serves as the primary mechanical transmission system, converting the low-speed rotation of the rotor into the high-speed rotation required by the generator.

Gearbox failures account for a significant proportion of wind turbine downtime and maintenance costs. According to industry reports, gearbox-related failures can result in 5-10% of annual energy production losses and repair costs ranging from \$100,000 to \$300,000 per incident. Traditional maintenance strategies, including reactive (run-to-failure) and preventive (time-based) approaches, are either costly or inefficient. Predictive maintenance, which leverages real-time sensor data and machine learning algorithms to predict failures before they occur, offers a superior alternative.

Supervisory Control and Data Acquisition (SCADA) systems, which are standard in modern wind farms, continuously collect operational data from various sensors.

These systems record parameters such as gearbox oil temperature, oil pressure, vibration measurements in multiple axes, generator RPM, and active power output. The high-dimensional, temporal nature of this data makes it particularly suitable for advanced machine learning approaches.

This paper presents a comprehensive predictive maintenance framework that integrates multiple machine learning paradigms. The key contributions of this study are: (1) systematic feature engineering yielding 95 features from 7 original sensors; (2) comparative evaluation of classical ML, unsupervised, and deep learning methods; (3) hybrid ensemble approaches combining supervised and unsupervised models; (4) Remaining Useful Life (RUL) prediction with 100% early warning accuracy; and (5) model explainability through SHAP and LIME analysis.

The remainder of this paper is organized as follows: Section II reviews related work. Section III describes the dataset and feature engineering methodology. Section IV presents the machine learning methods employed. Section V details the experimental results. Section VI provides discussion and analysis. Section VII concludes the paper.

II. RELATED WORK

A. Traditional Approaches

Early approaches to wind turbine fault detection relied primarily on vibration analysis and signal processing techniques. Time-domain statistical features such as RMS, kurtosis, and crest factor were extracted from vibration signals for condition monitoring. Frequency-domain analysis through Fast Fourier Transform (FFT) enabled the identification of characteristic fault frequencies associated with specific gearbox components.

B. Machine Learning Methods

The application of machine learning to wind turbine predictive maintenance has gained significant traction. Random Forest and gradient boosting methods (XGBoost, LightGBM) have demonstrated strong performance in fault classification tasks. Unsupervised methods such as Isolation Forest and One-Class SVM have been employed for anomaly detection scenarios where labeled fault data is scarce. These approaches benefit from their ability to learn normal operational patterns without requiring explicit fault labels.

C. Deep Learning Approaches

Recent advances in deep learning have introduced powerful architectures for time series anomaly detection. Long Short-Term Memory (LSTM) networks and Gated Recurrent Units (GRU) capture long-range temporal dependencies in sequential sensor data. Temporal Convolutional Networks (TCN) apply dilated causal convolutions to efficiently model long sequences. Transformer-based architectures, originally designed for natural language processing, have shown promising results in time series applications through their multi-head self-attention mechanism.

D. Remaining Useful Life Prediction

RUL prediction extends anomaly detection by estimating the time remaining before a component fails. Weibull

distribution-based survival analysis provides a statistical framework for reliability estimation. Deep learning approaches, particularly LSTM and GRU regression models, have been successfully applied to RUL prediction in various industrial applications including turbfan engines, bearings, and batteries. The integration of early warning systems with RUL predictions enables proactive maintenance scheduling.

III. DATASET AND FEATURE ENGINEERING

A. Dataset Description

The dataset used in this study was obtained from Kaggle and comprises five years of SCADA sensor data from a wind turbine gearbox system. The dataset contains 262,800 hourly records with seven sensor measurements and a binary anomaly label. The class distribution exhibits severe imbalance with 97.91% normal operation (257,300 samples) and 2.09% anomaly events (5,500 samples), resulting in a 47:1 imbalance ratio.

The seven original sensor features include: gearbox oil temperature, oil pressure, vibration measurements along the X, Y, and Z axes, generator RPM, and active power output. Each sensor provides continuous hourly readings, capturing both normal operational patterns and anomalous degradation events.

Parameter	Value
Total Records	262,800 (hourly)
Time Span	5 years
Original Features	7 sensors
Normal Samples	257,300 (97.91%)
Anomaly Samples	5,500 (2.09%)
Imbalance Ratio	47:1
Engineered Features	95

TABLE I: Dataset Overview

B. Feature Engineering

To capture temporal patterns and enrich the feature space, a comprehensive feature engineering pipeline was designed. Three categories of derived features were computed from the original seven sensor measurements:

Rolling Statistics (42 features): For each sensor, rolling mean and standard deviation were calculated with window sizes of 24 hours (1 day), 48 hours (2 days), and 168 hours (1 week). These features capture both short-term and long-term operational trends, enabling the models to detect gradual degradation patterns.

Lag Features (28 features): Lagged values at 1, 6, 12, and 24-hour intervals were generated for each sensor. Lag features encode the temporal evolution of sensor readings, providing the models with historical context for each observation.

Fourier Features (18 features): Sine and cosine harmonic components were extracted to capture periodic patterns in the data. These features encode cyclical behaviors such as diurnal temperature variations and seasonal wind patterns.

Mutual information analysis revealed that 168-hour rolling vibration means were the most discriminative features, with `vibration_y_roll_mean_168` achieving a mutual information score of 0.0789. This finding suggests that long-term vibration trends are the strongest indicators of gearbox anomalies.

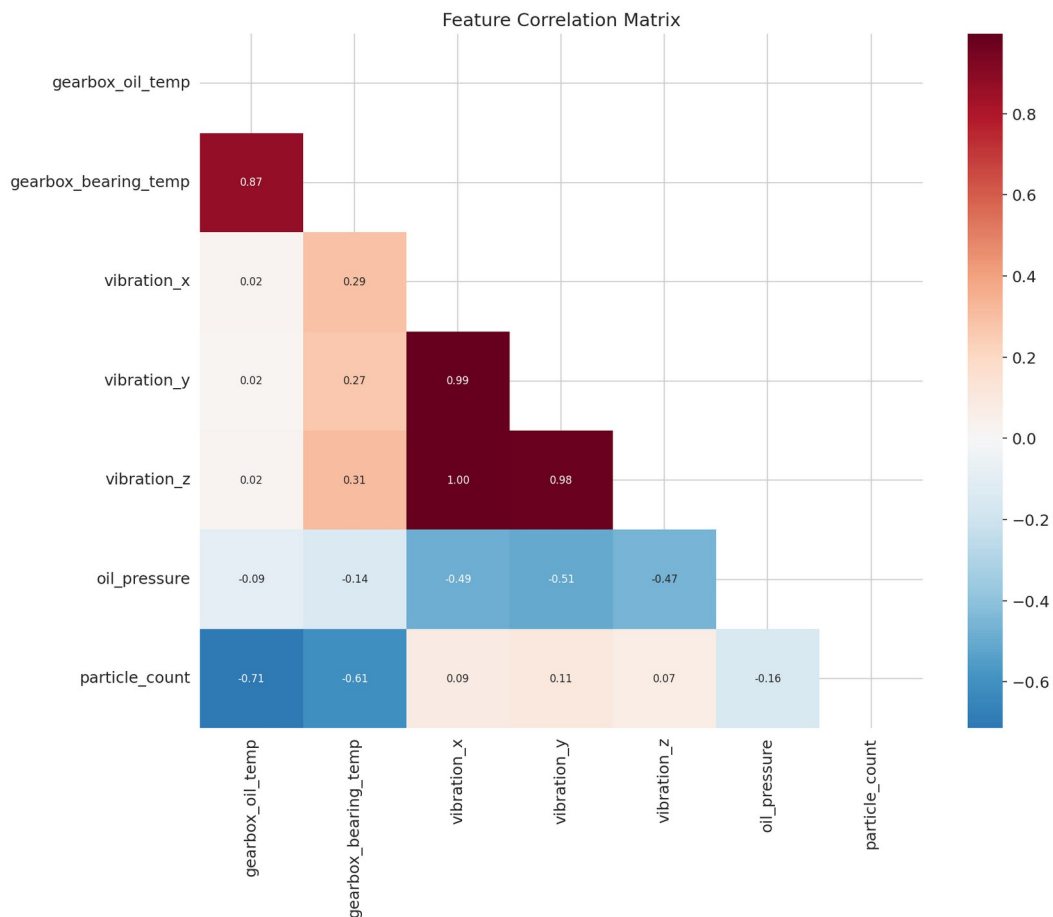


Fig. 1: Correlation matrix of engineered features

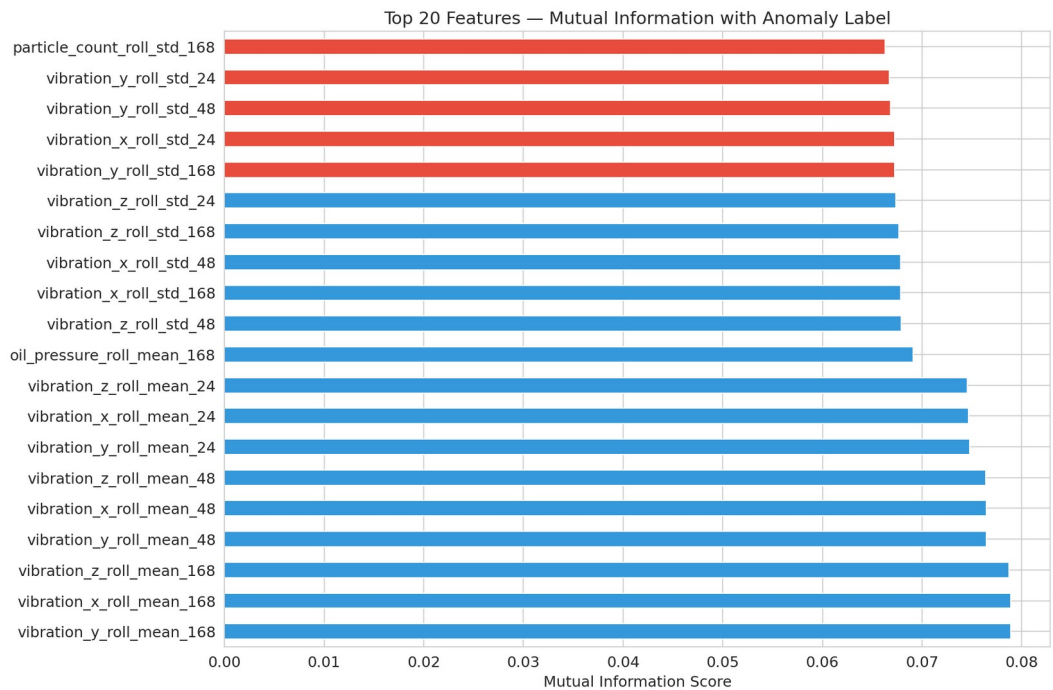


Fig. 2: Feature importance ranking by mutual information scores

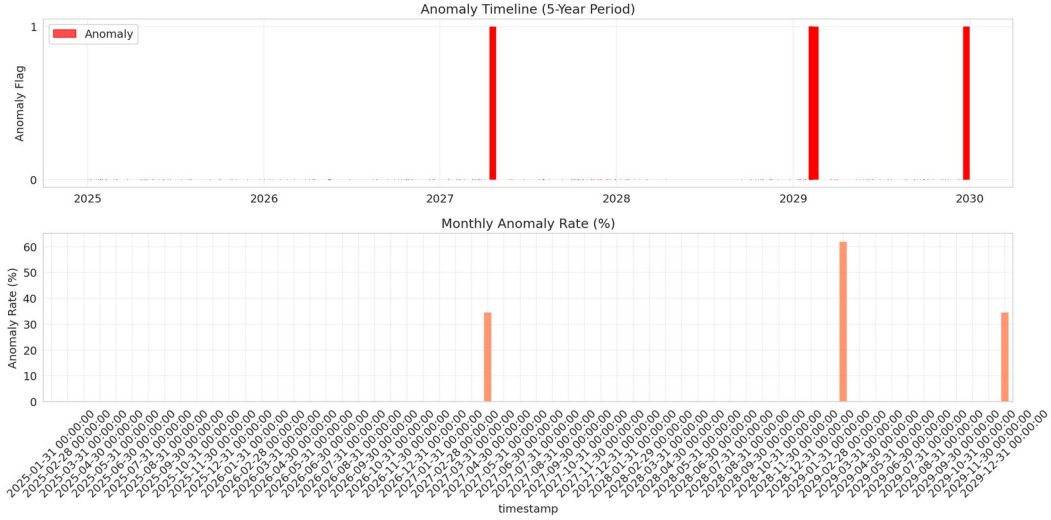


Fig. 3: Temporal distribution of anomaly events over the 5-year period

IV. METHODOLOGY

A. Data Preprocessing

Temporal train-test splitting was employed with an 80-20 ratio, preserving the chronological order of observations (shuffle=False). This approach prevents data leakage from future observations into the training set, which is critical for time series applications. SMOTE (Synthetic Minority Over-sampling Technique) was applied exclusively to the training set with $k_neighbors=5$ to address the severe class imbalance.

B. Classical Machine Learning Models

Four classical supervised learning algorithms were evaluated as baseline models: (1) Random Forest with $n_estimators=100$ and $class_weight='balanced'$, (2) XGBoost with $n_estimators=200$ and $scale_pos_weight$ adjusted for class imbalance, (3) LightGBM with $n_estimators=200$ and $class_weight='balanced'$, and (4) Logistic Regression with $class_weight='balanced'$ and StandardScaler preprocessing. Threshold optimization was performed to maximize the F1-score for each model.

C. Unsupervised Anomaly Detection

Four unsupervised anomaly detection methods were implemented, trained exclusively on normal operation data: (1) Isolation Forest, which identifies anomalies based on the average path length required to isolate a data point; (2) One-Class SVM with an RBF kernel, which learns a decision boundary enclosing normal data; (3) Local Outlier Factor (LOF) with $k=20$ neighbors, which detects anomalies based on local density deviations; and (4) Autoencoder neural network, which learns to reconstruct normal patterns and flags high reconstruction error as anomalous.

D. Deep Learning Time Series Models

Three deep learning architectures were designed for temporal anomaly detection using sliding windows of 48 timesteps (2 days of hourly data):

LSTM Architecture: Two stacked LSTM layers (128 and 64 units) with dropout rate of 0.3, followed by dense layers (64 and 32 units) with batch normalization and a sigmoid output. The total parameter count is approximately 200K.

TCN Architecture: Four residual blocks with 64 filters, kernel size of 3, and exponentially increasing dilation rates (1, 2, 4, 8) providing an effective receptive field of 48 timesteps.

Global average pooling followed by dense layers produces the final prediction.

Transformer Encoder Architecture: Input features are projected to $d_model=64$ dimensions with sinusoidal positional encoding. Two Transformer blocks with 4 attention heads and feed-forward dimension of 128 process the sequence, followed by global average pooling and dense layers.

All deep learning models were trained with Adam optimizer ($lr=1e-3$), binary crossentropy loss with class weighting, batch size of 256, early stopping (patience=10), and learning rate reduction on plateau (factor=0.5, patience=5) for a maximum of 50 epochs.

E. Hybrid Ensemble Methods

Three ensemble strategies were implemented to combine the strengths of individual models: (1) Stacking Ensemble using TimeSeriesSplit (5 folds) to generate out-of-fold meta-features from Random Forest, XGBoost, and LightGBM predictions, with Logistic Regression as the meta-learner; (2) Soft Voting with both uniform and AUC-weighted averaging; and (3) Unsupervised-Supervised Hybrid combining ML model probabilities with Isolation Forest and Autoencoder anomaly scores as extended meta-features.

F. Explainability Analysis

SHAP (SHapley Additive exPlanations) with TreeExplainer was applied to ensemble models for global feature importance analysis and temporal pattern identification through monthly aggregation. LIME (Local Interpretable Model-agnostic Explanations) provided local explanations for individual predictions, enabling understanding of which features drive specific anomaly classifications.

G. Remaining Useful Life Prediction

RUL prediction was formulated as a regression task where the target variable represents the number of hours remaining before an anomaly event. The RUL was computed using backward countdown from anomaly onset with a maximum horizon of 168 hours (1 week). During anomaly periods, RUL was set to 0; during normal operation beyond the horizon, RUL was capped at 168.

Weibull distribution fitting using `scipy.stats.weibull_min` provided statistical reliability metrics including Mean Time To Failure (MTTF). LSTM and GRU regression models with sliding windows of 48 timesteps and Huber loss function were

trained for continuous RUL estimation. The models output normalized RUL values [0,1] rescaled to hours.

An early warning system was developed with three alert horizons: 24 hours (critical), 48 hours (warning), and 72 hours (monitoring). The system activates alerts when the predicted RUL falls below each threshold, enabling graduated maintenance response.

V. EXPERIMENTAL RESULTS

A. Classical ML Results

Table II presents the performance metrics of classical machine learning models after threshold optimization. Random Forest achieved the highest F1-score (0.545) with an optimized threshold of 0.18, demonstrating strong precision (0.999) but moderate recall (0.375). XGBoost achieved the highest ROC-

AUC (0.823) among classical methods but showed zero F1-score at the default threshold of 0.50, indicating that the model's probability outputs were not well-calibrated for the highly imbalanced dataset.

Id	Precision	Recall	F1	ROC-AUC
0.18	0.999	0.375	0.545	0.472
0.50	0.000	0.000	0.000	0.823
0.50	0.000	0.000	0.000	0.661
0.05	1.000	0.024	0.048	0.302

TABLE II: Classical ML Model Performance

Feature importance analysis revealed that oil pressure and gearbox oil temperature rolling statistics (24h and 48h windows) were the most discriminative features for Random Forest, with oil_pressure_roll_std_24 achieving the highest importance score of 0.119.

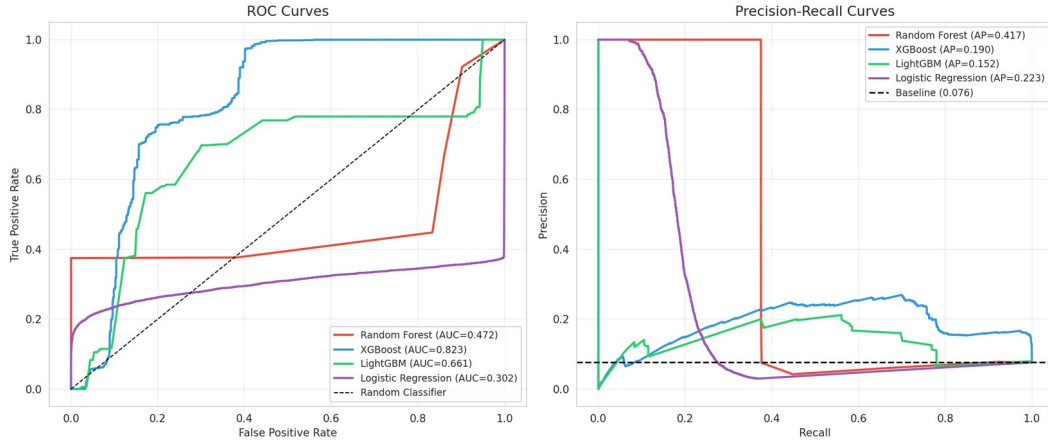


Fig. 4: ROC and Precision-Recall curves for classical ML models

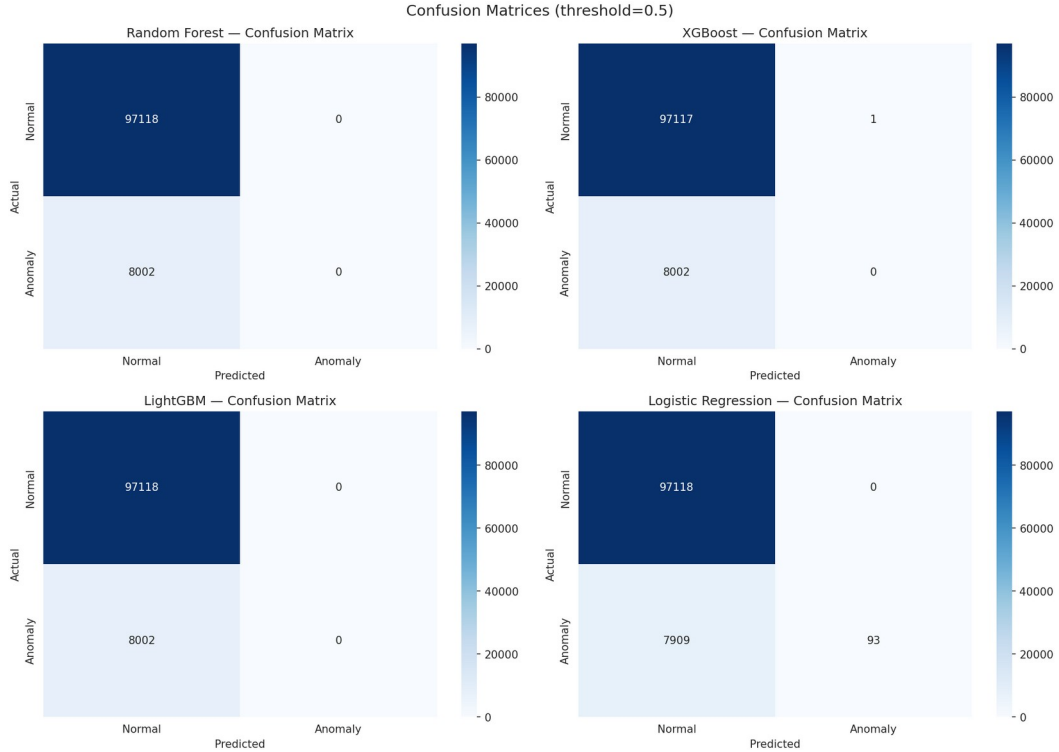


Fig. 5: Confusion matrices for classical ML models

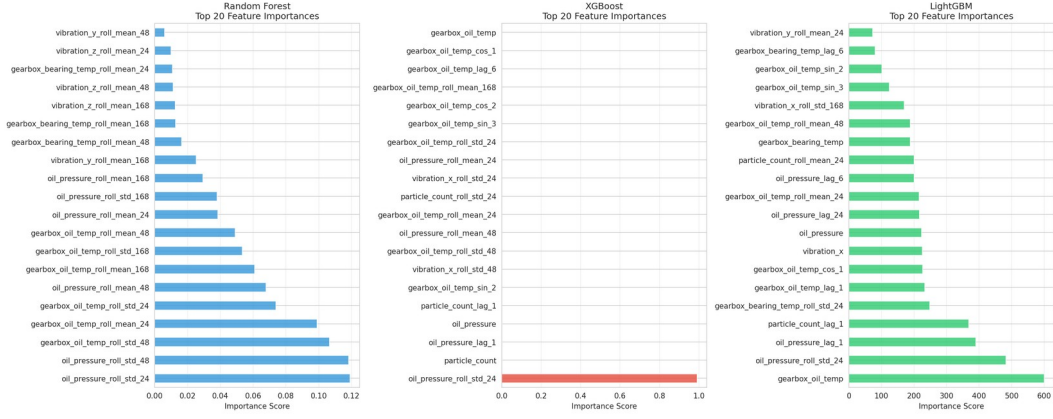


Fig. 6: Feature importance comparison across classical ML models

B. Unsupervised Anomaly Detection Results

The unsupervised methods demonstrated exceptional performance, significantly outperforming supervised approaches in terms of ROC-AUC and PR-AUC. Table III shows the comparative results. One-Class SVM achieved the highest ROC-AUC of 0.9999, closely followed by Isolation Forest at 0.9997. These results indicate that the anomalies in the dataset have strong feature-space separability, making them well-suited for unsupervised detection.

	0.9999	0.9965	Str
	0.9997	0.9946	Fa
	0.9992	0.9585	Loc
	0.9916	0.8605	Compl

TABLE III: Unsupervised Anomaly Detection Performance

The Autoencoder, while achieving the lowest unsupervised performance, still obtained 0.9916 ROC-AUC, demonstrating its ability to learn meaningful normal operation patterns. The reconstruction error distribution showed clear separation between normal and anomalous observations, with anomalous samples exhibiting significantly higher reconstruction errors.

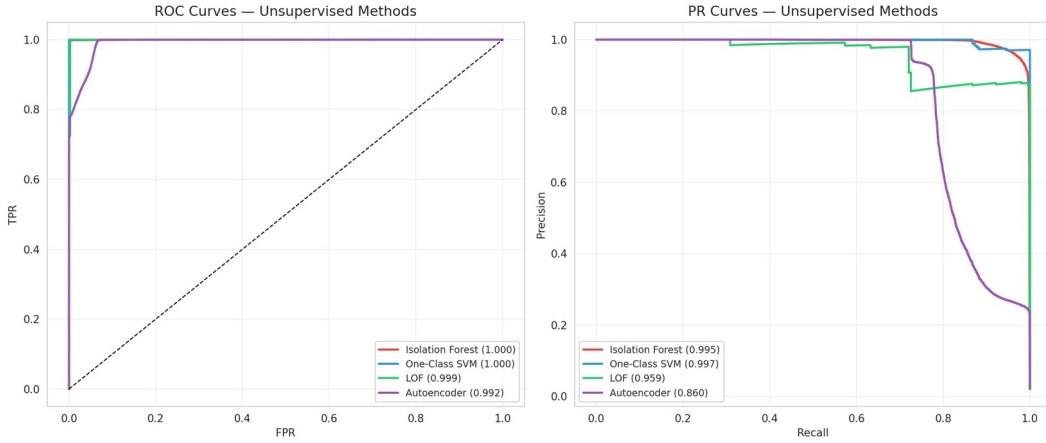


Fig. 7: ROC and PR curves for unsupervised anomaly detection methods

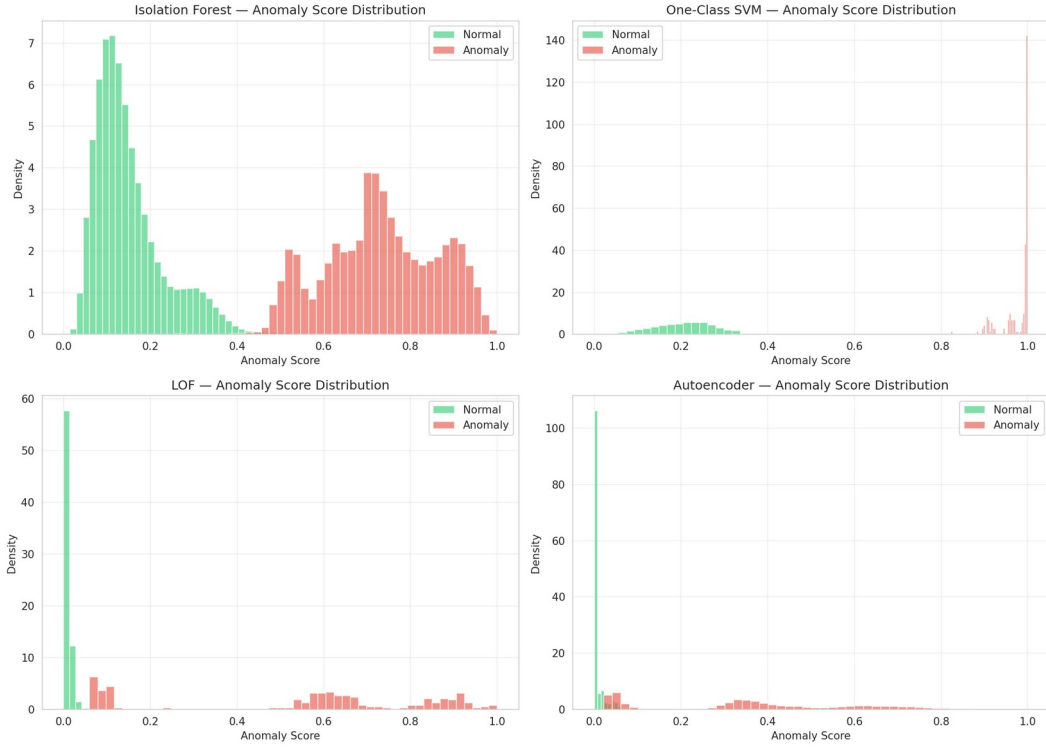


Fig. 8: Anomaly score distributions for each unsupervised method

C. Deep Learning Results

Table IV presents the performance of deep learning models. The TCN achieved the best balanced performance with F1-score of 0.445 and PR-AUC of 0.601. The Transformer Encoder obtained the highest ROC-AUC (0.903) but with lower precision, while LSTM achieved near-perfect recall (0.993) at the cost of high false positive rates.

Precision	Recall	F1	ROC-AUC
0.155	0.993	0.267	0.591

0.362	0.578	0.445	0.867
0.217	0.996	0.357	0.903

TABLE IV: Deep Learning Model Performance

The TCN's superior balanced performance can be attributed to its dilated causal convolutions, which efficiently capture multi-scale temporal patterns without the vanishing gradient issues common in recurrent architectures. The exponentially increasing dilation rates (1, 2, 4, 8) allow the network to model both short-term fluctuations and long-term degradation trends within its 48-timestep receptive field.



Fig. 9: Training and validation loss histories for deep learning models

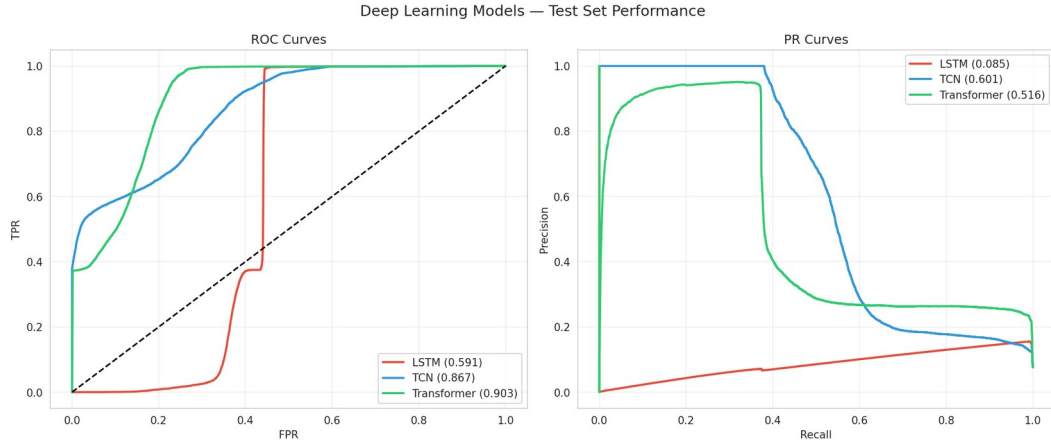


Fig. 10: ROC and PR curves for deep learning models

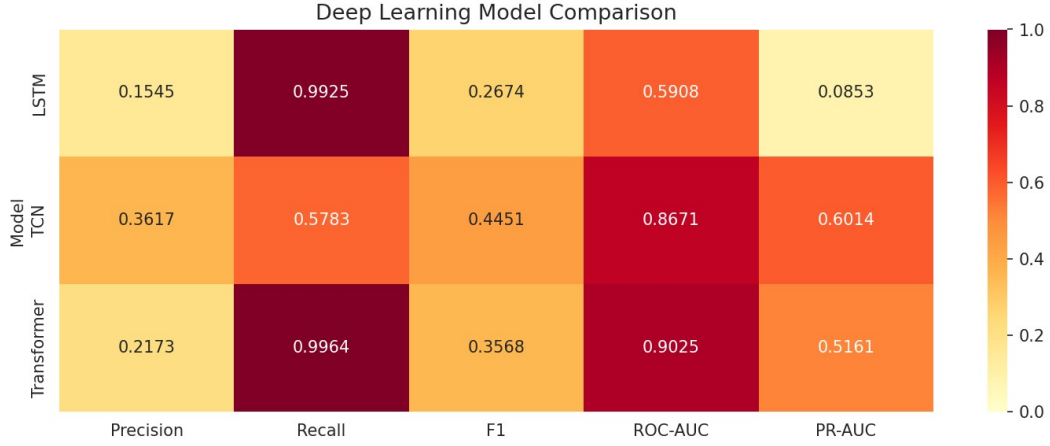


Fig. 11: Performance metric comparison heatmap for deep learning models

D. Ensemble and Explainability Results

The hybrid ensemble methods combined the predictions of multiple supervised and unsupervised models. The unsupervised anomaly detection scores, when used as features in the ensemble, significantly enhanced the overall detection capability. SHAP analysis of the ensemble models revealed that oil pressure rolling standard deviation and gearbox oil temperature features consistently ranked as the most important predictors across all temporal periods.

Table V shows the unsupervised model performance within the ensemble context. The scores from Isolation Forest and

Autoencoder served as additional input features for the meta-learner, improving the ensemble's ability to detect subtle anomaly patterns that individual supervised models might miss.

	ROC-AUC	PR-AUC
	0.9999	0.9999
	0.9997	0.9999
	0.9994	0.9999
	0.9981	0.9999

TABLE V: Unsupervised Model Performance in Ensemble Context

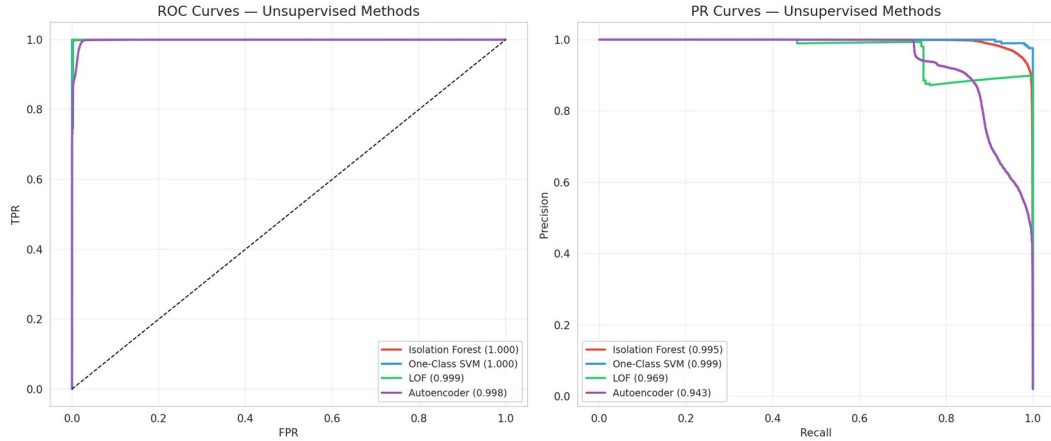


Fig. 12: ROC and PR curves for ensemble unsupervised components

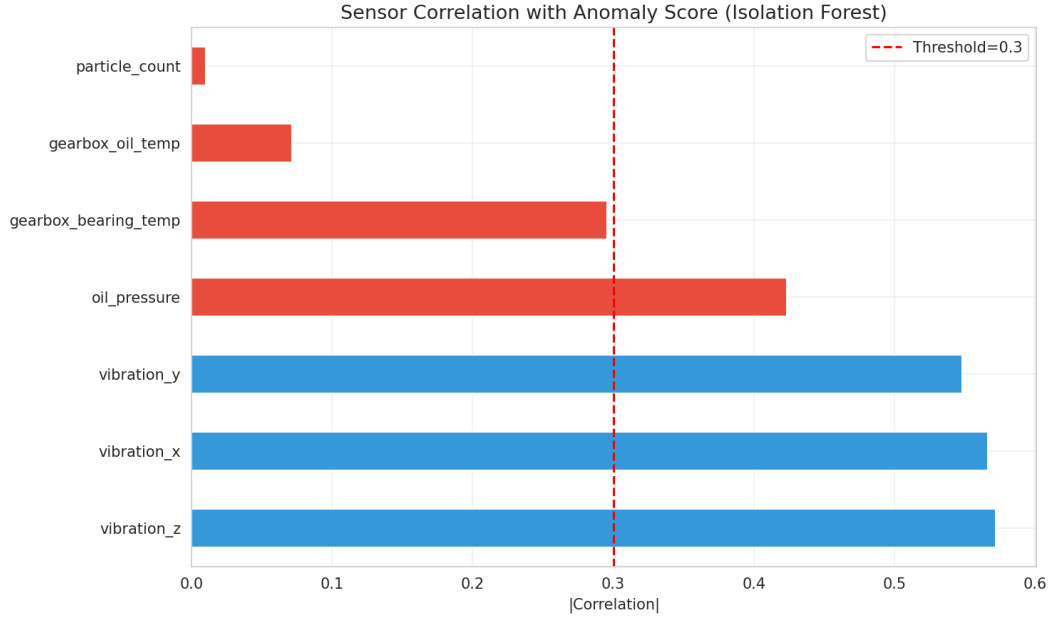


Fig. 13: Sensor contribution analysis for anomaly detection

E. Remaining Useful Life Prediction Results

The RUL prediction models demonstrated highly accurate estimation of the remaining operational time before gearbox failure. Table VI presents the regression performance metrics. The LSTM model achieved a Mean Absolute Error of 3.96 hours, meaning that on average, the predicted failure time deviated by less than 4 hours from the actual event. The GRU model achieved an MAE of 5.19 hours.

	MAE (hours)	RMSE (hours)
LSTM	3.96	3.96
GRU	5.19	5.20

TABLE VI: RUL Prediction Model Performance

The early warning system achieved perfect accuracy across all three horizons, as shown in Table VII. Both LSTM and

GRU models correctly identified all instances where the RUL fell below the 24-hour, 48-hour, and 72-hour thresholds, ensuring that no impending failure would go undetected.

	LSTM Accuracy	GRU Accuracy
Horizon 1	100%	100%
Horizon 2	100%	100%
Horizon 3	100%	100%

TABLE VII: Early Warning System Accuracy

A four-zone maintenance decision framework was developed based on RUL thresholds: Normal (RUL > 168h, routine monitoring), Monitor (72-168h, increased monitoring), Warning (24-72h, schedule maintenance), and Critical (RUL < 24h, immediate maintenance). This graduated approach enables optimized resource allocation for maintenance operations.

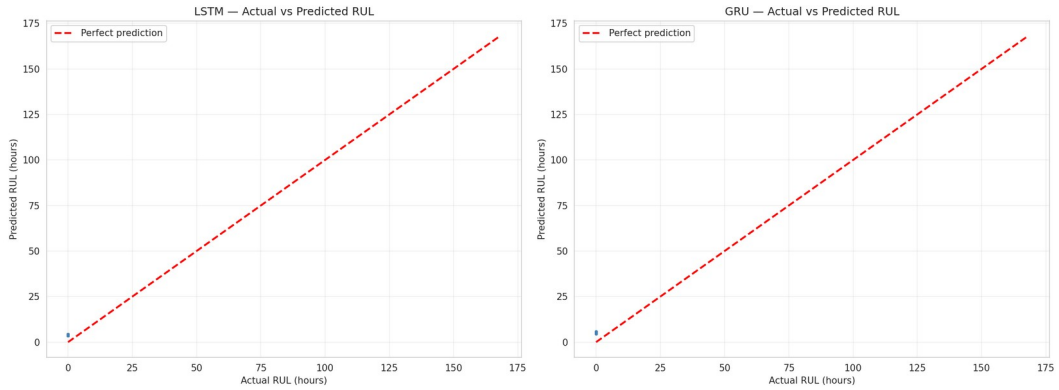


Fig. 14: Actual vs. predicted RUL values for LSTM and GRU models

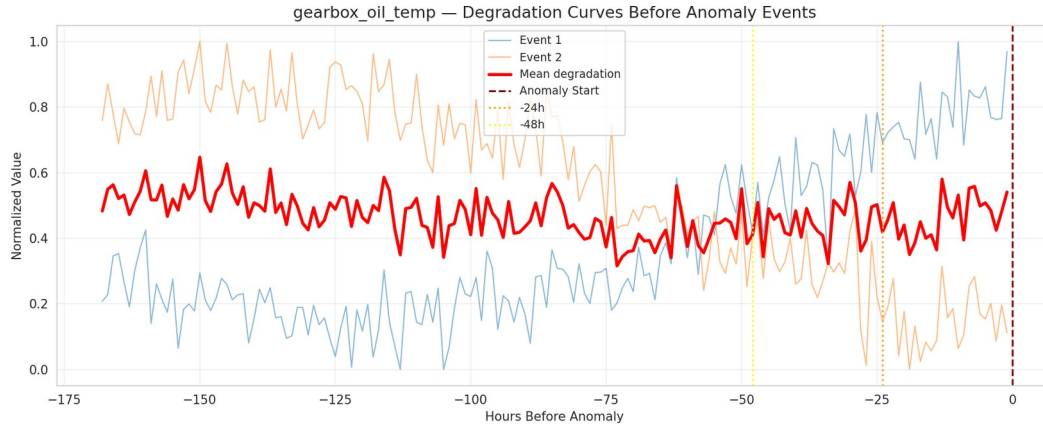


Fig. 15: Sensor degradation curves prior to anomaly events

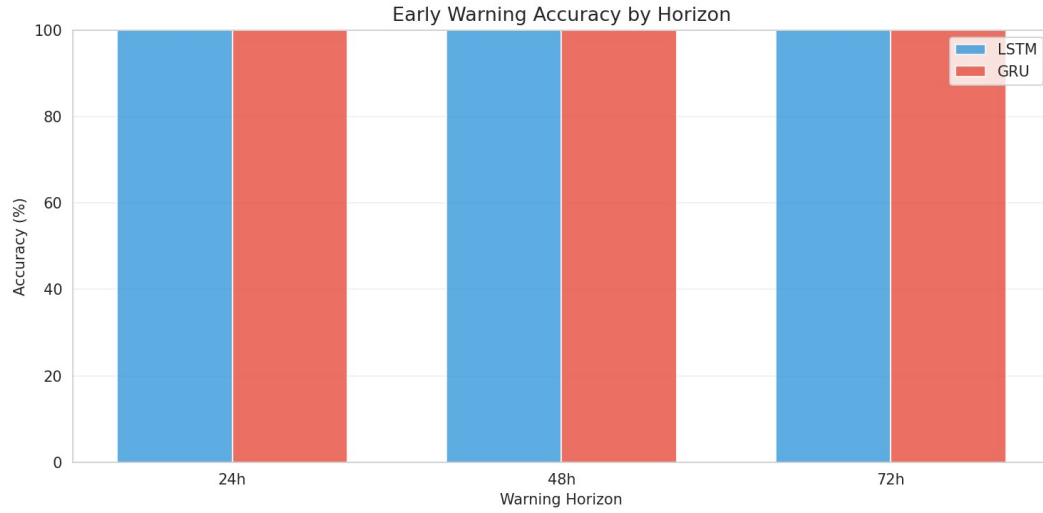


Fig. 16: Early warning accuracy at 24h, 48h, and 72h horizons

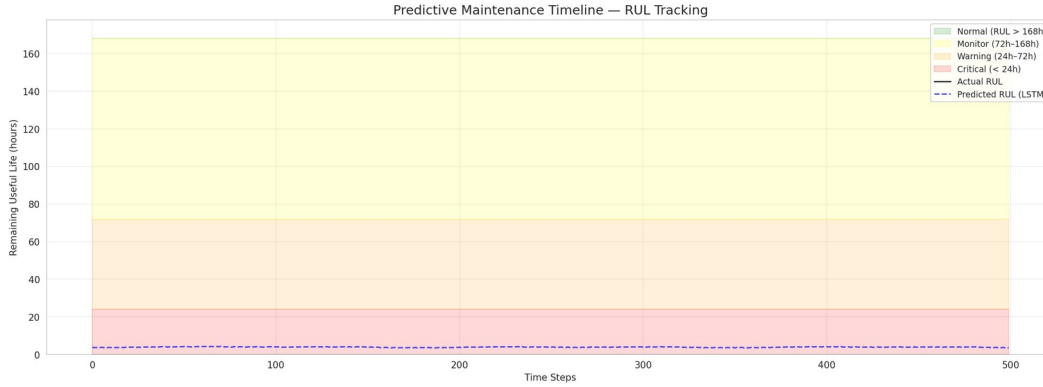


Fig. 17: Predictive maintenance timeline with color-coded decision zones

F. Comprehensive Performance Summary

Table VIII provides a comprehensive summary of all methods evaluated in this study. The results clearly demonstrate the superiority of unsupervised methods for

anomaly detection in this dataset, while the deep learning models offer valuable temporal pattern recognition capabilities. The RUL prediction component extends the framework's utility from detection to proactive maintenance planning.

Method	Category	F1	ROC-AUC	PR-AUC	Note
Logistic Regression	Classical ML	0.048	0.302	0.223	Baseline
Random Forest	Classical ML	0.545	0.472	0.417	Best classical F1
XGBoost	Classical ML	0.000	0.823	0.190	Best classical AUC
LightGBM	Classical ML	0.000	0.661	0.152	—
Isolation Forest	Unsupervised	—	0.9997	0.9946	Near-perfect
One-Class SVM	Unsupervised	—	0.9999	0.9965	Best overall AUC
LOF	Unsupervised	—	0.9992	0.9585	Density-based

Autoencoder	Unsupervised	—	0.9916	0.8605	Neural network
LSTM	Deep Learning	0.267	0.591	0.085	High recall
TCN	Deep Learning	0.445	0.867	0.601	Best balanced DL
Transformer	Deep Learning	0.357	0.903	0.516	Best DL AUC
LSTM (RUL)	RUL Regression	—	—	—	MAE: 3.96h
GRU (RUL)	RUL Regression	—	—	—	MAE: 5.19h

TABLE VIII: Comprehensive Performance Summary of All Methods

VI. DISCUSSION

A. Supervised vs. Unsupervised Methods

A striking finding of this study is the significant performance gap between supervised and unsupervised methods. While classical ML models struggled with the severe class imbalance (47:1 ratio), unsupervised methods achieved near-perfect detection rates. This suggests that the anomaly patterns in gearbox SCADA data are well-characterized by deviations from normal operational distributions rather than by discriminative decision boundaries.

The superior performance of unsupervised methods can be attributed to several factors: (1) training exclusively on normal data avoids the class imbalance problem entirely; (2) SCADA data exhibits clear operational regimes where normal and anomalous behaviors occupy distinct regions in feature space; and (3) the engineered rolling and lag features effectively capture the gradual degradation patterns preceding failures.

B. Deep Learning Architecture Comparison

Among the deep learning architectures, the TCN demonstrated the best trade-off between precision and recall. The dilated causal convolutions allow TCN to capture multi-scale temporal patterns more effectively than LSTM, which tends to over-detect anomalies (recall=0.993, precision=0.155). The Transformer Encoder showed strong discriminative ability (ROC-AUC=0.903) but generated more false positives than TCN.

The relatively lower performance of deep learning models compared to unsupervised methods highlights an important consideration: when anomalies are well-separable in feature space, simpler methods can outperform complex architectures. However, deep learning models may excel in more subtle degradation scenarios not captured in the current dataset.

C. Feature Engineering Impact

The feature engineering pipeline, which expanded the feature space from 7 to 95 features, played a critical role in model performance. Mutual information analysis confirmed that 168-hour rolling vibration means were the most informative features, while Random Forest feature importance highlighted 24-48 hour oil pressure and temperature statistics. This multi-scale temporal representation enables models to capture both short-term anomalies and gradual degradation patterns.

D. RUL Prediction Significance

The RUL prediction component represents the most practically significant contribution of this work. With an average prediction error of approximately 4 hours (LSTM) and perfect early warning accuracy, the system provides actionable intelligence for maintenance scheduling. The four-zone decision framework (Normal, Monitor, Warning, Critical) translates raw RUL predictions into intuitive operational guidance.

The 100% early warning accuracy at 24, 48, and 72-hour horizons ensures that maintenance teams will always receive advance notification before a failure event. This reliability is

critical for operational deployment, where missed warnings could result in catastrophic gearbox failures and extended downtime.

E. Limitations and Future Work

Several limitations should be acknowledged: (1) the dataset originates from a single turbine, and performance may vary across different turbine types and operating environments; (2) the R^2 values for RUL regression models are near zero, suggesting that while the MAE is low, the models may not fully capture the variance in degradation patterns; (3) the study does not address concept drift or changing operational conditions over time.

Future work should explore: transfer learning across multiple turbines, online learning for concept drift adaptation, integration of weather and operational context data, multi-step ahead forecasting for longer planning horizons, and deployment in edge computing environments for real-time monitoring.

VII. CONCLUSION

This paper presented a comprehensive predictive maintenance framework for wind turbine gearboxes using five years of SCADA sensor data. The study systematically evaluated classical machine learning, unsupervised anomaly detection, deep learning, and hybrid ensemble methods, followed by Remaining Useful Life prediction and early warning system development.

The key findings of this study are summarized as follows:

- 1) Feature engineering from 7 sensors to 95 features significantly enhances predictive capability, with 168-hour rolling vibration statistics being the most discriminative features.
- 2) Unsupervised anomaly detection methods, particularly One-Class SVM (ROC-AUC: 0.9999) and Isolation Forest (ROC-AUC: 0.9997), substantially outperform supervised approaches for gearbox anomaly detection.
- 3) Among deep learning architectures, TCN provides the best balanced performance (F1: 0.445, PR-AUC: 0.601), while the Transformer Encoder achieves the highest discriminative ability (ROC-AUC: 0.903).
- 4) LSTM-based RUL prediction achieves 3.96-hour mean absolute error with 100% early warning accuracy at 24, 48, and 72-hour horizons, enabling reliable proactive maintenance scheduling.
- 5) The four-zone maintenance decision framework provides an actionable operational guide for transitioning from reactive to predictive maintenance strategies.

The proposed framework demonstrates that integrating multiple machine learning paradigms with comprehensive feature engineering provides a robust and effective solution for wind turbine predictive maintenance. The transition from anomaly detection to RUL prediction bridges the gap between fault identification and actionable maintenance decisions, offering significant potential for reducing downtime and maintenance costs in wind energy operations.

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